



Timeseries Statistics Model

Statistical Analysis of Hydrological Timeseries Data

MKM Research Labs

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1 Executive Summary

This document describes the Timeseries Statistics Model developed by MKM Research Labs for the quantitative analysis of hydrological timeseries data. The model provides a unified statistical engine that operates on both synthetic water level records (generated by the gauge historical daily simulator) and observed river flow data sourced from the National River Flow Archive (NRFA).

The model computes a comprehensive suite of descriptive, distributional, and extreme-event statistics from daily observation records. Its outputs serve three primary functions within the Physical Risk framework: (i) characterising the baseline hydrological regime at each gauge for risk reports, (ii) computing flood exceedance frequencies against Environment Agency thresholds to support hazard assessment, and (iii) extracting annual maxima series that feed into the GEV Hazard Curve Model for return period estimation.

The implementation is a single-function pipeline with two convenience wrappers that handle field-name remapping for backward compatibility with existing JSON schemas. All computations are performed in pure Python using the standard library `statistics` module, with no external numerical dependencies.

2 Model Purpose and Scope

2.1 Objective

The Timeseries Statistics Model serves as the descriptive analytics layer for hydrological data within the Physical Risk framework. Its primary objectives are:

- (i) Compute basic descriptive statistics (mean, median, standard deviation, min, max) from daily timeseries observations.
- (ii) Calculate a full percentile distribution (P5 through P99) to characterise the empirical value distribution.
- (iii) Count flood threshold exceedances at three Environment Agency severity levels (Flood Alert, Flood Warning, Severe Flood Warning) and derive annualised exceedance frequencies.
- (iv) Extract monthly mean values to capture seasonal hydrological patterns.
- (v) Compute annual maxima series with dates and rank the top extreme events for hazard analysis.
- (vi) Identify extreme events (values above the 99th percentile) for tail-risk reporting.

2.2 Scope

The model is designed to process daily-resolution timeseries data from two sources:

- **Synthetic water levels:** Generated by the gauge historical daily (`gaugehd`) simulator, with values in the field `level_meters`. These are synthetic stochastic series representing plausible daily water levels at each gauge, used when observed data are unavailable.
- **NRFA river flows:** Observed daily mean flow data from the National River Flow Archive, with values in the field `flow_cumecs` (cubic metres per second).

The model is agnostic to the data source; it operates on a configurable value key, allowing it to process any numeric timeseries with a date field. Flood threshold exceedance analysis is only performed when flood stage thresholds are provided, which is applicable to water level data but not typically to flow data.

2.3 Regulatory Context

The statistics produced by this model are used in gauge risk reports and property-level flood assessments. While the model itself is primarily descriptive, its outputs—particularly flood exceedance frequencies and annual maxima—are inputs to the GEV Hazard Curve Model, which is subject to bank model validation requirements under PRA SS1/23. Accurate and auditable computation of these statistics is therefore a prerequisite for downstream model validation.

3 Mathematical Framework

3.1 Descriptive Statistics

Given a timeseries of n daily observations $\{x_1, x_2, \dots, x_n\}$ with corresponding dates $\{d_1, d_2, \dots, d_n\}$, the model computes:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad (1)$$

$$\tilde{x} = \text{median}(x_1, \dots, x_n), \quad (2)$$

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}, \quad (3)$$

where $\bar{x}$ is the sample mean, $\tilde{x}$ is the sample median, and s is the sample standard deviation (Bessel-corrected). The record length in years is computed as $T = n/365.25$.

3.2 Percentile Computation

The empirical percentile function uses a nearest-rank method. For percentile $p \in \{5, 10, 25, 50, 75, 90, 95, 99\}$, the percentile value is:

$$Q(p) = x_{(\lceil n \cdot p / 100 \rceil)},$$

where $x_{(k)}$ denotes the k -th order statistic of the sorted sample. This provides a non-parametric characterisation of the empirical distribution without assuming any particular distributional form.

For NRFA flow data, the hydrological flow quantiles Q5 and Q95 are derived by mapping to the complementary percentiles: Q95 (the flow exceeded 95% of the time) corresponds to the 5th percentile, and Q5 (the flow exceeded 5% of the time) corresponds to the 95th percentile:

$$Q95_{\text{flow}} = Q(5), \quad (4)$$

$$Q5_{\text{flow}} = Q(95). \quad (5)$$

3.3 Flood Exceedance Analysis

For water level data where flood stage thresholds $\{\theta_{\text{alert}}, \theta_{\text{warning}}, \theta_{\text{severe}}\}$ are provided, the model counts the number of daily observations exceeding each threshold:

$$N_{\theta} = \sum_{i=1}^n \mathbf{1}(x_i \geq \theta),$$

where $\mathbf{1}(\cdot)$ is the indicator function. The annualised exceedance frequency is:

$$f_{\theta} = \frac{N_{\theta}}{T} = \frac{N_{\theta} \times 365.25}{n}.$$

This provides an empirical estimate of how many days per year the water level at a gauge exceeds each flood severity threshold.

3.4 Monthly Mean Decomposition

The timeseries is partitioned by calendar month $m \in \{01, 02, \dots, 12\}$, and the mean is computed within each partition:

$$\bar{x}_m = \frac{1}{n_m} \sum_{i:\text{month}(d_i)=m} x_i,$$

where n_m is the number of observations in month m . This captures the seasonal cycle in water levels or flows, which is driven by rainfall seasonality, evapotranspiration patterns, and catchment storage dynamics.

3.5 Annual Maxima Extraction

The annual maximum series is constructed by extracting the largest observation in each calendar year y :

$$M_y = \max_{i:\text{year}(d_i)=y} x_i.$$

The date of each annual maximum is recorded, and the top 10 events are ranked in descending order. Summary statistics of the annual maxima series—mean $\bar{M}$ and standard deviation s_M —provide a first-order characterisation of the extreme value behaviour before formal GEV fitting.

3.6 Extreme Event Detection

Extreme events are defined as observations exceeding the 99th percentile threshold $Q(99)$:

$$\mathcal{E} = \{(d_i, x_i) : x_i \geq Q(99)\}.$$

The count $|\mathcal{E}|$ and the top 10 events by magnitude are reported. By construction, approximately 1% of all observations qualify as extreme events, but their temporal clustering and magnitude provide important context for tail-risk assessment.

4 Input Parameters and Calibration

4.1 Observation Data

The model requires a list of daily observations, each containing a date string (ISO 8601 format, YYYY-MM-DD) and a numeric value field. The value field name is configurable:

Table 1: Observation input fields.

Field	Type	Description
date	String	Observation date in YYYY-MM-DD format
level_meters	Float	Daily water level in metres (synthetic data)
flow_cumecs	Float	Daily mean river flow in m ³ /s (NRFA data)

4.2 Flood Stage Thresholds

Flood exceedance analysis requires three thresholds, sourced from the gauge CDM:

Table 2: Flood stage threshold parameters.

Parameter	Key	Description
Flood Alert	FloodAlert	Level at which flooding is possible
Flood Warning	FloodWarning	Level at which flooding is expected
Severe Flood Warning	SevereFloodWarning	Level indicating danger to life

Thresholds are optional. When omitted (e.g. for flow data), the flood exceedance section is skipped.

4.3 Calibration

The model has no free parameters requiring calibration. All computations are deterministic given the input data. The only configurable parameter is `value_key`, which selects the numeric field to analyse. Convenience wrappers (`calculate_level_statistics` and `calculate_flow_statistics`) set this automatically and handle backward-compatible key remapping.

5 Implementation Details

5.1 Source Files

The model is implemented in a single Python module:

Table 3: Source file inventory.

File	Responsibility
<code>models/statistics/timeseries.py</code>	Core function <code>calculate_timeseries_statistics()</code> and two convenience wrappers: <code>calculate_level_statistics()</code> for water level data and <code>calculate_flow_statistics()</code> for NRFA flow data.

5.2 Function Architecture

The implementation follows a single-pass pipeline within `calculate_timeseries_statistics()`:

1. **Value extraction:** Extract numeric values and dates from the observation list using the configured `value_key`.
2. **Basic statistics:** Compute mean, median, standard deviation, min, max, and record the date of the maximum value.
3. **Percentiles:** Sort the values and compute P5 through P99 using the nearest-rank method.
4. **Flood exceedances:** If flood stage thresholds are provided, count observations above each threshold and compute annualised frequencies.
5. **Flow quantiles:** For flow data (`flow_cumecs`), map P5→Q95 and P95→Q5.
6. **Monthly means:** Partition observations by calendar month and compute per-month averages.
7. **Annual maxima:** Partition by calendar year, extract the maximum value and its date per year, and rank the top 10 events.
8. **Extreme events:** Identify observations exceeding the P99 threshold and return the top 10 by magnitude.

5.3 Convenience Wrappers

Two thin wrappers handle field-name remapping for backward compatibility with existing JSON schemas:

- `calculate_level_statistics(observations, flood_stages)`: Calls the core function with `value_key="level_meters"`, then remaps all output keys from `level_meters` to `level`.
- `calculate_flow_statistics(observations)`: Calls the core function with `value_key="flow_cumecs"` (no flood stages), then remaps output keys from `flow_cumecs` to `flow`.

5.4 Dependencies

The model uses only the Python standard library **statistics** module for mean, median, and standard deviation computations. No external numerical libraries (NumPy, SciPy) are required. All rounding is to 3 decimal places unless otherwise specified.

6 Sensitivity Analysis

The Timeseries Statistics Model is purely descriptive and deterministic—there are no estimated parameters or model assumptions that introduce sensitivity in the traditional sense. However, the outputs are sensitive to properties of the input data:

Table 4: Flood Alert exceedance sensitivity to threshold level (20-year synthetic record).

Alert Threshold (m)	Total Days	Days/Year
2.50	1120	56.0
2.75	361	18.1
3.00	68	3.40
3.25	10	0.5000
3.50	0	0.0000

Table 5: Summary statistics from 20-year synthetic water level record.

Statistic	Value (m)
Mean level	1.998
Median level	1.998
Std dev	0.466
Min level	0.647
Max level	3.434
P5	1.246
P95	2.747
P99	2.994
Annual maxima mean	3.239
Annual maxima std	0.140

6.1 Record Length

The reliability of all computed statistics depends on the length of the observation record. Key sensitivities include:

- **Percentiles:** The P99 estimate requires at least 100 observations to be meaningful. For records shorter than 5 years ($\sim 1,825$ observations), the P99 estimate is based on fewer than 20 data points and may be unstable.
- **Annual maxima:** The number of annual maxima equals the number of complete years in the record. Fewer than 10 years of data produces an annual maxima series too short for reliable extreme value analysis downstream.
- **Monthly means:** Seasonal decomposition requires at least one complete year of data. Partial years at the start or end of the record may bias monthly means if certain months are under-represented.

6.2 Flood Threshold Values

Exceedance counts are highly sensitive to the flood stage thresholds. A 10% reduction in the Flood Alert threshold can increase the annualised exceedance frequency by 50–100%, depending on the steepness of the water level distribution in the threshold region. Accurate threshold data from the Environment Agency are therefore critical.

6.3 Data Quality

The model does not perform outlier detection or data quality checks. Erroneous observations (e.g. sensor spikes, missing-data fill values) will directly affect all computed statistics. In particular:

- A single anomalous high value will inflate the maximum, P99, and annual maxima series.
- Gaps filled with zeros or default values will depress the mean and lower percentiles.

Data quality screening should be performed upstream before statistics are computed.

7 Validation and Backtesting

7.1 Cross-Check Against NRFA Published Statistics

For gauges with NRFA records, the model's computed statistics (mean flow, Q5, Q95) are compared against the NRFA's published gauging station summary statistics. Agreement within 5% confirms correct implementation of the core statistical calculations.

7.2 Synthetic Data Validation

A suite of unit tests validates the model against synthetic datasets with known properties:

- **Constant series:** A timeseries with all values equal to a constant c should yield mean = c , standard deviation = 0, and all percentiles = c .
- **Linear ramp:** A monotonically increasing series of length n should produce P50 equal to the midpoint value, and the annual maximum for each year should equal the last observation of that year.
- **Known exceedances:** A series with exactly k values above a threshold θ should produce an exceedance count of k and an annualised frequency of k/T .

7.3 Consistency Checks

The following internal consistency conditions are verified:

- $\min \leq P5 \leq P25 \leq P50 \leq P75 \leq P95 \leq \max$.
- The sum of monthly observation counts equals the total observation count.
- The number of annual maxima equals the number of distinct years in the record.
- Flood exceedance counts satisfy: $N_{\text{severe}} \leq N_{\text{warning}} \leq N_{\text{alert}}$.

8 Model Limitations and Known Weaknesses

8.1 Nearest-Rank Percentile Method

The nearest-rank method used for percentile computation is the simplest of several standard methods (linear interpolation, Hazen, Weibull plotting position, etc.). For small samples, it can produce step-function behaviour in the percentile estimates. More sophisticated interpolation methods (e.g. the default method in NumPy's `percentile` function) would produce smoother estimates but were not adopted to avoid external dependencies.

8.2 No Data Quality Screening

The model processes all observations as provided, with no detection of outliers, sensor failures, missing-data sentinels, or temporal gaps. Users are responsible for ensuring data quality before invoking the statistics pipeline.

8.3 Calendar Year Partitioning

Annual maxima are extracted by calendar year (January–December), which may not align with the hydrological year (typically October–September in the UK). This could split a single flood season across two calendar years, potentially underestimating the true annual maximum if a major event spans the year boundary.

8.4 No Uncertainty Quantification

The model reports point estimates only; no confidence intervals or standard errors are provided for the computed statistics. For short records, sampling uncertainty may be substantial, particularly for extreme percentiles and annual maxima statistics.

8.5 Flat Monthly Structure

Monthly means are computed across all years, which masks inter-annual variability. A wet year and a dry year contribute equally to the monthly mean for a given month. The model does not report monthly standard deviations or inter-annual ranges.

9 Governance Validation Questions

The following nine questions are mandated by Chapter 5 of the *Model Risk Governance for Vendors* handbook and must be explicitly addressed for every model in the inventory. Response status is tracked in the Model Governance Dashboard (MRC portal).

9.1 Q1 — Ronseal Test

“Does the model do what it says on the tin?”

Yes. The model computes a comprehensive suite of statistics from daily hydrological timeseries, delivering: (i) descriptive statistics (mean, median, standard deviation, min, max), (ii) full percentile distribution (P5–P99), (iii) flood threshold exceedance counts and annualised frequencies at three EA severity levels, (iv) monthly mean values for seasonal characterisation, (v) annual maxima series with dates and ranking, and (vi) extreme event identification (above 99th percentile). All documented outputs are produced.

9.2 Q2 — Accuracy and Stability

“Does the model provide timely, accurate and stable results?”

The model is fully deterministic: identical timeseries inputs produce identical statistical outputs. All computations use well-defined mathematical operations (sample mean, sample standard deviation with Bessel correction, order statistics for percentiles). The implementation uses the Python standard library `statistics` module with no external numerical dependencies. Processing of a 10-year daily timeseries (3,652 observations) completes in under 50 ms.

9.3 Q3 — Resource Usage

“Does the model consume excessive computational resources?”

No. The model is a single-function pipeline with $O(n \log n)$ complexity (dominated by the sort required for percentile computation, where n is the number of observations). For a 30-year record ($n \approx 11,000$), processing completes in under 100 ms. Memory usage is $O(n)$. The model has no external dependencies beyond the Python standard library. Batch processing of all gauges in a catchment completes in under 1 second.

9.4 Q4 — Override Controls

“Are there controls over manual overrides?”

No manual override mechanism exists. The model computes pure statistical summaries from input data with no tuneable parameters. Flood threshold exceedance analysis uses EA-defined thresholds provided in the gauge configuration data, which are not modifiable by the model. The percentile computation uses a fixed methodology (simple order-statistic indexing). The annual maxima extraction algorithm is deterministic and parameter-free.

9.5 Q5 — Output Consumption

“How is the output consumed for risk decision-making?”

Statistical outputs serve three primary decision-support functions: (a) gauge risk reports displaying baseline hydrological regime characterisation for MRC review; (b) flood exceedance frequencies informing hazard assessment and the GEV Hazard Curve Model (MKM-HAZARD-001); and (c) annual maxima series feeding directly into GEV distribution fitting for return period estimation. Monthly means support seasonal flood risk monitoring. Extreme event identification triggers ad-hoc review.

9.6 Q6 — Model Chaining

“Is the model’s output used as input to another model?”

Yes. The model’s primary downstream consumer is the GEV Hazard Curve Model (MKM-HAZARD-001), which uses the annual maxima series for extreme value distribution fitting. Flood exceedance frequencies are consumed by gauge-level risk reports and the governance dashboard. Upstream, the model processes data from two sources: (i) the synthetic gauge historical daily (`gaugehd`) simulator, and (ii) observed NRFA river flow records. The model has no upstream model dependencies—it operates directly on data.

9.7 Q7 — Weaknesses

“Does the model have weaknesses and limitations of use?”

Yes; the principal limitations are documented in Section 8. Key weaknesses: (i) simple order-statistic percentile method (may differ from interpolation-based methods at extreme percentiles); (ii) annualised exceedance frequency assumes uniform distribution of events across the record (may be inaccurate for short records); (iii) annual maxima extraction requires at least one year of data; (iv) no trend detection or non-stationarity testing; (v) no data quality checks for gaps, outliers, or measurement errors.

9.8 Q8 — Monitoring

“What type of monitoring is required?”

Minimal ongoing monitoring is required given the purely descriptive nature of the model. Recommended checks: (a) annual verification that computed statistics for NRFA gauges are consistent with NRFA published statistics; (b) review of any changes to the Python `statistics` module implementation across version upgrades; (c) validation that flood exceedance counts are consistent with EA published flood event records for monitored gauges; (d) data completeness check (record length in years) before relying on annualised frequencies.

9.9 Q9 — Overall Risk

“Is the overall model risk acceptable?”

Acceptable. As a Tier 3 descriptive analytics model, the Timeseries Statistics Engine carries low inherent model risk. It applies well-defined statistical operations with no calibration, tuneable parameters, or modelling assumptions beyond standard statistical methodology. The primary risk is data quality (garbage in, garbage out), which is mitigated by the record length reporting and extreme event flagging. The dependency-free implementation ensures portability and auditability.

10 Change History

11 Document History

Date	Version	Description	Author
13-Feb-2026	1.0	Initial documentation of Timeseries Statistics Model	D. Kelly
18-Feb-2026	1.1	Added Section 9: Governance Validation Questions (Chapter 5)	D. Kelly